

Tharun Kumar Marada

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PROFESSIONAL SUMMARY

MSc Computer Science graduate with two machine learning internships. My focus is model evaluation: I build the test data, choose the metrics, run the benchmark and say plainly which model is better and which is not. Recent work covers a thesis benchmarking 9 classifiers on human and LLM-written text, plus LLM agents and document question answering in Python.

TECHNICAL SKILLS

Evaluation | Benchmark design, test set building, precision, recall, F1, AUC, Precision@k, grouped cross-validation, data leakage checks, robustness testing, grid search tuning

AI and LLMs | Prompt design, parsed structured outputs, RAG, LangGraph, LangChain, ChromaDB, Hugging Face sentence embeddings (Sentence-BERT), Groq API (LLaMA 3)

Data and Tools | Python, SQL, Pandas, NumPy, scikit-learn, XGBoost, Matplotlib, Seaborn, Flask, Streamlit, Git
Languages | English (Fluent), Swedish (Basic), Telugu (Native), Hindi (Intermediate)

EDUCATION

MSc Computer Science Blekinge Institute of Technology, Sweden Aug 2024 – Jun 2026

- Thesis: detected AI-generated and paraphrased text with 90.3% accuracy and 0.94 AUC by benchmarking 9 classifiers on 21 embedding stability features
- Built the evaluation set by cleaning 85,449 HC3 samples down to 74,974 and drawing 5,000 balanced human and ChatGPT texts
- Tested robustness on 25,000 paraphrased, edited and truncated variants generated with LLaMA 3.1
- Showed that a KNN baseline on static embeddings falls to 61.6% accuracy while the same model on stability features holds 89.3%

BTech Computer Science and Engineering JNTU, India 2020 – 2024

WORK EXPERIENCE

Machine Learning Intern FlyRank.ai Jul 2026 – Sep 2026

- Kept the reported scores free of data leakage by validating with client-grouped splits, so that no client appeared in both training and test data
- Reached 0.90 Precision@20 and 0.70 Precision@50 by building a Random Forest model in Python, covering data preparation, feature engineering, training and evaluation
- Made results reproducible by building data preparation, transformation and validation as one repeatable workflow

Machine Learning Intern BIST Technologies Summer 2025

- Delivered anomaly detection and classification models on operational data, keeping outputs reliable by adding automated data quality checks at each stage
- Turned model findings into recommendations for non-technical stakeholders by presenting results in plain language

KEY PROJECTS

Predictive Maintenance Model Benchmark [LINK](#)

- Chose Random Forest (92% precision, 79% F1) over Isolation Forest (34% F1) and DBSCAN (7% F1) by benchmarking all three on 10,000 sensor readings with a 3.4% failure rate
- Reported DBSCAN as not ready for production after finding its 100% recall came with only 3% precision

Document Question Answering Chatbot (RAG) [LINK](#)

- Answered questions on a 187-page annual report with cited answers in under 3 seconds by building a RAG pipeline with LangChain, ChromaDB and LLaMA 3.1
- Guarded against made-up answers by designing a prompt that makes the model cite figures, name the report section and say when the answer is not in the document
- Retrieved the 5 most relevant passages per question by embedding 1,000-character chunks with a Hugging Face sentence model (all-MiniLM-L6-v2)

Multi-Agent AI Research Generator [LINK](#)

- Produced cited research reports in under 90 seconds by building a 4-agent LangGraph pipeline (planner, searcher, extractor, writer) that draws on 15 to 20 live sources
- Kept the pipeline running on malformed model output by having the planner return a parsed list of subtopics with a fallback

LLM-Based Intrusion Detection System [LINK](#)

- Reached 99.73% F1 and a 0.43% false positive rate in 5-fold cross-validation on a public benchmark by training a Random Forest on 25,000 network connections
- Explained each alert in plain English by passing the prediction and its confidence to LLaMA 3